

Physics Packet: Christmas & New Year's Review

Name: _____

Date: _____

1) Central Park Ice Skating

Bob is at Wollman Rink in Central Park, getting ready to ice skate. Bob has a mass of $m = 35$ kg). The coefficient of kinetic friction on the ice is $\mu_k = 0.04$. When he gets on the rink, he pushes himself off to start skating with a push force of $F = 25$ N horizontally.

1. Draw a FBD.

2. Find the force of gravity F_g .

3. Find the normal force F_N .

4. Find the kinetic friction force F_f .

5. Find $\sum F_x$.

6. Find $\sum F_y$.

7. Find the acceleration a .

2) Santa's Sled

Santa is ready to deliver all the presents to the good children all around the world. Santa's sled with all the presents has a mass $m = 1500$ kg. It slides on level snow in the North Pole with a coefficient of kinetic friction $\mu_k = 0.12$. His sled is pulled by his reindeer at a force of 3500 N.

1. Draw a FBD.
2. Find the force of gravity F_g .
3. Find the normal force F_N .
4. Find the kinetic friction force F_f .
5. Find $\sum F_x$.
6. Find $\sum F_y$.
7. Find the acceleration a .

3) Reindeer Pulling Santa's Sleigh

Santa's reindeer pull his sleigh which has a mass $m = 500$ kg sleigh on level ground with a rope tension $T = 140$ N at an angle $\theta = 25^\circ$ above horizontal.

1. Draw a FBD.
2. Find the force of gravity F_g .
3. Find F_{Tx} .
4. Find F_{Ty} .
5. Find $\sum F_x$.
6. Find $\sum F_y$.
7. Find the acceleration a .

4) Mont Tremblant Ski Resort

A skier has a mass $m = 70$ kg and is skiing down a black diamond angled at $\theta = 65^\circ$.

1. Draw a FBD.

2. Find the force of gravity F_g .

3. Find the normal force F_N .

4. Find F_{gx} .

5. Find F_{gy} .

6. Find $\sum F_x$.

7. Find $\sum F_y$.

8. Find the acceleration a .

5) Ornament Pulley

A $m = 8.0$ kg box of ornaments hangs at rest from two identical vertical ropes sharing the load equally (two ropes straight up, no angles).

1. Draw a FBD.
2. Find the force of gravity F_g .
3. Find the tension force F_T .
4. Find $\sum F_x$.
5. Find $\sum F_y$.
6. Find the acceleration a .

6) New Year's Eve Ball pt 1.

The Times Square Ball Drop ball has mass $m = 1200$ kg. Before New Year's, it is brought up with an acceleration of $a = 0.15$ m/s².

1. Draw a FBD.

2. Find the force of gravity F_g .

3. Find the tension force F_T .

4. Find $\sum F_x$.

5. Find $\sum F_y$.

7) New Year's Eve Ball pt 2.

The same ball is now lowered straight down at constant speed using a cable.

1. Draw a FBD.

2. Find the force of gravity F_g .

3. Find the tension force F_T .

4. Find $\sum F_x$.

5. Find $\sum F_y$.

8) New Year's Eve Ball pt 3.

The same ball is now lowered with an acceleration of $a = 0.30 \text{ m/s}^2$.

1. Draw a FBD.

2. Find the force of gravity F_g .

3. Find the tension force F_T .

4. Find $\sum F_x$.

5. Find $\sum F_y$.

9) Pulling a bag of gifts

A gift bag of mass $m = 20$ kg rests on frictionless, level ground. Two elves pull on the bag with ropes at different angles: Rope A pulls with force $F_{TA} = 40$ N at 60° above the horizontal (horizontally to the right). Rope B pulls with force $F_{TB} = 30$ N at 30° above the horizontal (horizontally to the left).

1. Draw a FBD.
2. Find the force of gravity F_g .
3. Find the tension force F_{TxA} .
4. Find the tension force F_{TyA} .
5. Find the tension force F_{TxB} .
6. Find the tension force F_{TyB} .
7. Find $\sum F_x$.

8. Find $\sum F_y$.

9. Determine the magnitude and direction of the resultant force acting on the bag.

10. Find the acceleration a .

10) Snow Tube Tow

A snow tube with a rider has mass $m = 45$ kg and is pulled across frictionless, level snow by a tow rope with tension $T = 160$ N at an angle $\theta = 30^\circ$ above the horizontal.

1. Draw a FBD.

2. Find the force of gravity F_g .

3. Find the tension force F_{Tx} .

4. Find the tension force F_{Ty} .

5. Find the tension force F_T .

6. Find $\sum F_x$.

7. Find $\sum F_y$.

8. Determine the magnitude and direction of the resultant force acting on the bag.

9. Find the acceleration a .